Your Language Model Can Secretly Write Like Humans: Contrastive Paraphrase Attacks on LLM-Generated Text Detectors

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Abstract

The misuse of large language models (LLMs), such as academic plagiarism, has driven the development of detectors to identify LLM-generated texts. To bypass these detectors, paraphrase attacks have emerged to purposely rewrite these texts to evade detection. Despite the success, existing methods require substantial data and computational budgets to train a specialized paraphraser, and their attack efficacy greatly reduces when faced with advanced detection algorithms. To address this, we propose **Co**ntrastive **P**araphrase **A**ttack (CoPA), a training-free method that effectively deceives text detectors using off-the-shelf LLMs. The first step is to carefully craft instructions that encourage LLMs to produce more human-like texts. Nonetheless, we observe that the inherent statistical biases of LLMs can still result in some generated texts carrying certain machine-like attributes that can be captured by detectors. To overcome this, CoPA constructs an auxiliary machine-like word distribution as a contrast to the human-like distribution generated by the LLM. By subtracting the machine-like patterns from the human-like distribution during the decoding process, CoPA is able to produce sentences that are less discernible by text detectors. Our theoretical analysis suggests the superiority of the proposed attack. Extensive experiments validate the effectiveness of CoPA in fooling text detectors across various scenarios. The code is available at: https://github.com/ffhibnese/CoPA_Contrastive_Paraphrase_Attacks

1 Introduction

Large language models (LLMs), such as GPT-4 and Claude-3.5, have demonstrated remarkable abilities in text comprehension and coherent text generation. These capabilities have driven their widespread applications, including code generation (Jiang et al., 2024) and academic research (Stokel-Walker, 2022). However, the misuse of LLMs for harmful purposes, such as academic plagiarism and misinformation generation, has raised significant societal concerns regarding safety and ethics (Bao et al., 2024; Wu et al., 2025; Kong et al., 2025). In response, various detection methods that leverage the unique characteristics of LLM-generated texts from multiple perspectives have been proposed to mitigate the associated risks.

Concurrently, red-teaming countermeasures (Krishna et al., 2023; Shi et al., 2024) have also been introduced to evaluate the reliability of these detection algorithms, which can be broadly categorized into *word-substitution* and *paraphrase* attacks. Specifically, word-substitution attacks (Shi et al., 2024; Wang et al., 2024) replace specific important words in the fake sentence using candi-

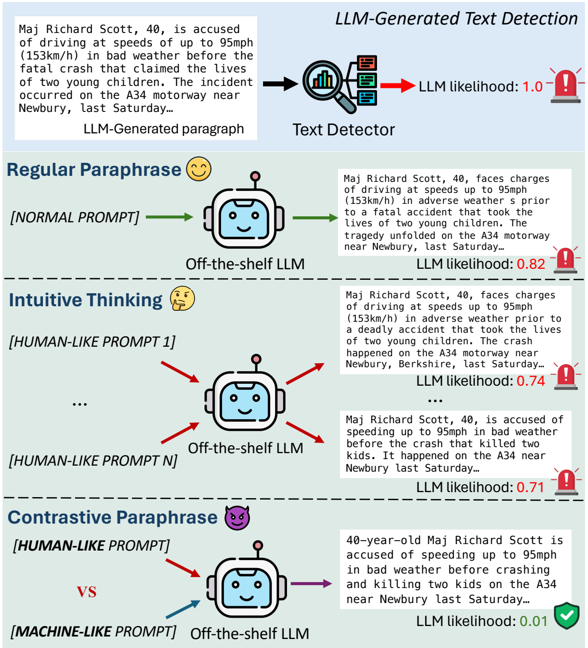

Figure 1: Comparison of different paraphrasing strategies. The human-like and machine-like prompts are crafted to guide the LLM in generating human-style and machine-style texts, respectively.

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date words generated by a language model. However, they require an additional surrogate model to identify the word importance, and the replacement operation can significantly increase sentence perplexity (Wang et al., 2024), making the processed texts easily identified by humans. In contrast, Dipper (Krishna et al., 2023) proposes a paraphrase-based attack that rewrites the whole paragraph by modifying the syntax and phrasing to deceive text detectors. This approach does not rely on surrogate models and can preserve sentence perplexity, presenting a more practical and versatile attack strategy. Nonetheless, Dipper requires training a large-scale generative language model as the paraphraser, incurring substantial data collection and computational burdens. Moreover, the attack performance is greatly diminished when confronted with more advanced defense strategies such as Fast-DetectGPT (Bao et al., 2024), as shown Sec. 4.2.

In this paper, we build upon the research line of paraphrase attacks and propose a training-free paraphrase approach named Contrastive Paraphrase Attack (CoPA), which aims to elicit human-written word distributions from an off-the-shelf LLM to evade detection. Specifically, we revisit the fundamental mechanisms underlying existing detection algorithms and hypothesize that the core principle of effective paraphrase attacks lies in erasing the machine-inherent characteristics within the paragraph while incorporating more human-style features, such as more flexible choices of words and phrases. Based on this insight, we intuitively seek to craft prompts that alleviate the established statistical constraints of pre-trained LLMs and produce more human-like word distributions, as shown in Fig. 1. While this strategy exhibits some effectiveness, LLMs pretrained on massive corpora hold a strong tendency to prioritize words with high statistical probability to ensure sentence coherence (Bao et al., 2024; Mao et al., 2024). This inherent bias consistently influences the word choices of generated sentences, irrespective of the input prompts. Therefore, some paraphrased sentences still exhibit certain machine-related characteristics, rendering them highly discernible by detection classifiers.

To address this limitation, we conduct a reverse-thinking analysis. While it is challenging to directly produce highly human-written word distributions that can fully bypass detection, eliciting the opposite machine-like word distributions that contain rich machine-related attributes is considerably easier. These machine-like word probabilities can then serve as negative instances to further purify the obtained human-style distribution for more human-like text generation. In light of this consideration, an auxiliary machine-like distribution is constructed as a contrastive reference, which is used to filter out the machine-related concepts from the aforementioned human-like distribution. By sampling from the meticulously adjusted word distribution, CoPA generates more diverse and human-like sentences, which exhibit remarkable effectiveness in deceiving LLM-text detectors.

Contributions. We propose CoPA, a novel paraphrase attack that contrastively modifies the word distribution from an off-the-shelf LLM to rewrite generated texts for enhanced attacks against text detectors. CoPA eliminates the cumbersome burdens of training a dedicated paraphraser, achieving an efficient and effective attack paradigm. Furthermore, we develop a theoretical framework that substantiates the superiority of CoPA.

To validate the effectiveness, we conduct extensive experiments on 3 long-text datasets with various styles against 8 powerful detection algorithms. Compared to baselines, CoPA consistently enhances the attack while maintaining semantic similarity, *e.g.*, an average improvement of 57.72% in fooling rates (at FPR=5%) for texts generated by GPT-3.5-turbo when against Fast-DetectGPT.

2 Related Work

2.1 LLM-generated Text Detection

Existing detection algorithms for AI-generated text can generally be categorized into two types: (i) Training-based detection: These methods typically involve training a binary classification language model. Specifically, OpenAI employs a RoBERTa model (Liu, 2019) trained on a collection of millions of texts for detection. To enhance the detection robustness, RADAR (Hu et al., 2023) draws inspiration from GANs (Goodfellow et al., 2020) and incorporates adversarial training between a paraphraser and a detector. Additionally, DeTective (Guo et al., 2024a) proposes a contrastive learning framework to train the encoder to distinguish various writing styles of texts, and combined with a pre-encoding embedding database for classification. R-detect (Song et al., 2025) employs a kernel relative test to judge a text by determining whether its distribution is closer to that of human texts. Despite the efforts in specific domains, training-based methods are struggling with generalization to unseen language domains, which reduces their practicality and versatility. (ii) Zero-shot detection: These methods are training-free and typically focus on extracting inherent features of LLMs' texts to make decisions. GLTR (Gehrmann et al., 2019) and LogRank(Solaiman et al., 2019) leverage the probability or rank of the next token for detection. Since AI-generated text typically exhibits a higher probability than its perturbed version, DetectGPT (Mitchell et al., 2023) proposes the probability curvature to distinguish LLM and human-written text. Building on this, Fast-DetectGPT (Bao et al., 2024) greatly improves the efficiency by introducing conditional probability curvature that substitutes the perturbation step with a more efficient sampling step. TOCSIN (Ma and Wang, 2024) presents a plug-and-play module that incorporates random token deletion and semantic difference measurement to bolster zero-shot detection capabilities. Other methods explore different characteristics, such as likelihood (Hashimoto et al., 2019), N-gram divergence (Yang et al., 2024), and the editing distance from the paraphrased version (Mao et al., 2024).

2.2 Attacks against Text Detectors

Red-teaming countermeasures have been proposed to stress-test the reliability of detection systems. Early attempts evade detection using in-context learning (Lu et al., 2023) or directly fine-tuning the LLM (Nicks et al., 2023) under a surrogate detector. Recent advances can be broadly categorized into:

Substitution-based attacks. Shi et al. (2024) introduce the substitution-based approach, which minimizes the detection score provided by a surrogate detector by replacing certain words in AI sentences with several synonyms generated by an auxiliary LLM. Subsequently, RAFT (Wang et al., 2024) improves the attack performance by introducing an LLM-based scoring model to greedily identify critical words in the machine sentences. However, this type of attack relies on an additional surrogate model, and the generated sentences suffer from reduced coherence and fluency, which are easily identifiable by human observations and limit their practical utility in real-world scenarios.

Paraphrasing-based attacks. Conversely, Dipper (Krishna et al., 2023) suggests a surrogate-free approach that can perfectly maintain text perplexity. With a rewritten dataset of paragraphs with altered word and sentence orders, Dipper fine-tunes a T5-XXL (Raffel et al., 2020) as a paraphraser to rewrite entire machine paragraphs, which effectively fools text detectors while preserving semantic consistency. Based on Dipper, Sadasivan et al. (2023) introduce a recursive strategy that performs multiple iterations of paraphrasing, with slightly degrading text quality while significantly enhancing attack performance. Raidar (Mao et al., 2024) proposes a straightforward approach that directly queries an LLM to paraphrase machine text into paragraphs with more human-style characteristics. Shi et al. (2024) suggest a strategy that automatically searches prompts to induce more human-like LLM generations. However, it relies on a strong assumption that a surrogate detector is available.

This paper follows the more practical and applicable paradigm of surrogate-free paraphrasing attacks and proposes a contrastive paraphrasing strategy, which achieves remarkable efficacy in bypassing LLM-text detection systems.

3 Method

In this section, we first present the paradigm of paraphrase attacks. Then, we elaborate on the proposed CoPA that rewrites generated texts using a pre-trained LLM. Finally, we provide a theoretical framework to guarantee the attack effectiveness.

3.1 Problem Formulation

We denote the AI-generated text detector weighted by w as $D_w : \mathcal{Y} \to [0, 1]$, where $\mathcal{Y}$ denotes the text domain. The detector D_w maps text sequences $y \in \mathcal{Y}$ to corresponding LLM likelihood scores, where higher values indicate a greater probability of being LLM-generated. For an LLM pretrained on extensive corpora, the resulting texts exhibit significant writing styles, including word preferences and coherent syntactic structures, which have been leveraged in previous studies to develop various text detectors. To evade detection, a malicious paraphrase attacker aims to reduce these machine-related concepts within the machine-generated texts to obtain paraphrased variants that can effectively mislead D_w into outputting lower LLM likelihood scores. Before delving into the proposed method, we first review the token generation paradigm of the LLM inference. We consider utilizing an off-the-shelf LLM as the paraphraser. Given an input instruction x, a machine text y_m and a pre-trained LLM $f_\theta(\cdot)$ parameterized by θ, f_θ generates the paraphrased paragraph y by producing tokens in an autoregressive manner. At each timestep t, $f_\theta(\cdot)$ samples the next token y_t from the